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FINALE PRODUCT PITCH

TEAM: HACK O’ GIANTS

DATE: 10/11/2025

Hack'O Giants

Team member details

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Product pitch

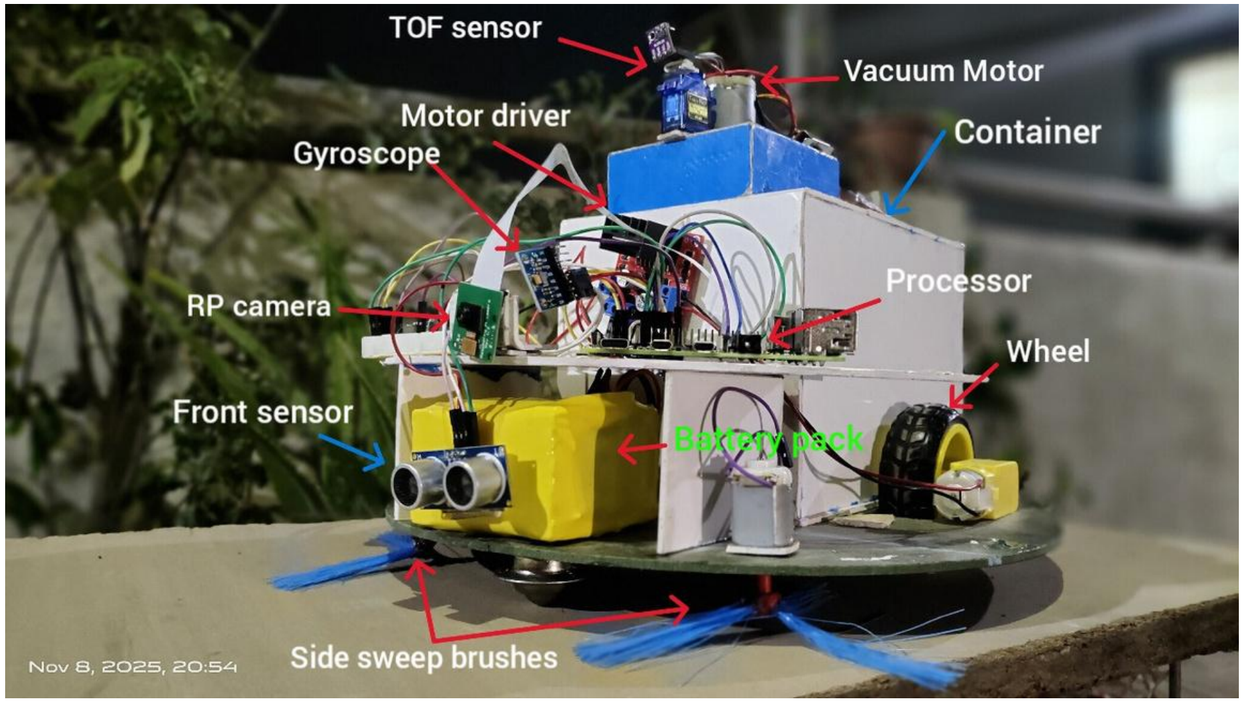
CHOTU – “Compact Home Operations & Tidying Unit”

Overview:

CHOTU is an autonomous floor-cleaning robot designed for smart homes and offices. Powered by Raspberry Pi 4 and integrated with AI-based obstacle detection (YOLOv8), it ensures efficient, adaptive cleaning across varied environments.

Key Highlights:

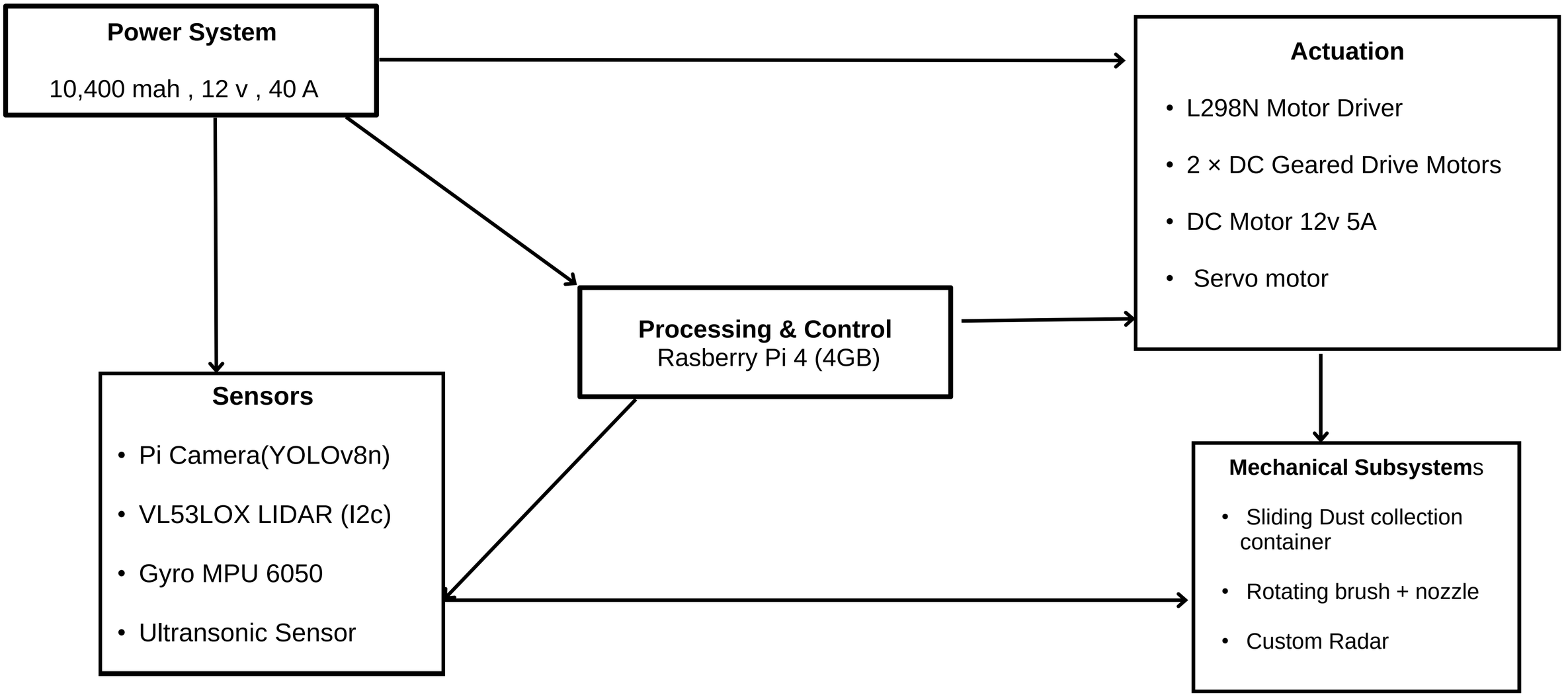
- **AI Vision System**: Real-time object detection using YOLOv8 to identify and avoid obstacles like chairs, tables, and toys.
- **Smart Navigation**: Zig-zag cleaning pattern with encoder feedback for complete area coverage.
- **Dual Cleaning Mechanism**: Rotating brushes and high-speed vacuum motor for effective dry waste pickup.
- **Energy Efficient**: 10,400 mAh Li-ion battery (12V, 40A) ensures up to 60 minutes of runtime per charge.
- **Compact & Modular**: Designed for 10×12 ft rooms, easy maintenance, and component scalability.



Tech Stack

- **Hardware**: Raspberry Pi 4, TOF Sensor (VL53L0X), ToF sensors, gyroscope, Encoders, Pi Camera,Vacuum Motor,Wheel,Ultrasonic Front sensor,Motor Driver.
- **Software**: Python, OpenCV, Ultralytics YOLOv8

Block diagram



Key technical specs & features

Core Specs:

- **Processor**: Raspberry Pi 4 (4 GB RAM, Quad Core 1.5 GHz)
- **AI Engine**: YOLOv8n (Real-time Object Detection)
- **Motors**: 2 × 12 V DC Geared (10 RPM) + Bergmann Tornado Motor
- **Motor Driver**: L298N PWM Controller
- **Sensors**: LiDAR dTOF (VL53L0X), Gyro Sensor Vibration sensor, Ultrasonic sensor, Pi Camera
- **Battery**: Dual Li-Po (3S4P 10,400 mAh, 12 V, 40 A)
- **Chassis**: MDF board + Acrylic sheet (≈ 2.8 kg)
- **Runtime**: 35–45 min | Recharge: 2–3 hrs

Key Features:

- **AI-Based Vision:** YOLO object detection for obstacle classification
- **Dual Action Cleaning:** Brush + Vacuum mechanism for dry waste

Differentiators :

AI-Driven Autonomy | Dual Power System | Modular Design

Battery calculations

Formula:

Runtime_min = (Battery_mAh / 1000) / I_total × 60

Given Data:
Battery: 11.1 V / 10 Ah (111 Wh)
I_total = I_pi + I_vl + I_ultra + I_motors + I_driver
I_total = 2.5 + 0.05 + 0.02 + 2.0 + 0.05 = 4.62 A

Runtime Calculation:

Runtime = 10 / 4.6 = 2.17 h ≈ 130 minutes

Summary:

Average Load Current: ≈ 4.6 A
Expected Runtime: ≈ 2.1 h (130 min)
Power Draw: 11.1 V × 4.6 A = 51.1 W
Efficiency: 80–85% (≈ 100 min actual runtime)

Bill of Materials

Item Description	Quantity	Rate (per unit)	CGST	SGST	Amount
BT-65 - 11.1V 18650 LITHIUM	1	300	9	9	300
ELCOM - RSF-11-RSF-1104-7-04-	2	15	9	9	30
E-79 SW-420 VIBRATION TILT	1	40	9	9	40
BO motor Wheel Encoder Disc	2	260	46.8	46.8	520
VL53L0X TOF Based LIDAR Laser	3	110	29.7	29.7	330
MPU-6050 Module 3 Axis Gyroscope Accelerometer	1	140	12.6	12.6	140
Raspberry Pi 5MP Camera Module	1	180	16.2	16.2	180
1kg 2025 PLA PRO 3D Printing	1	720	64.8	64.8	720
Bo Moter	2	50	-	-	100
IRF520	1	47	-	-	47
Single holder	1	25	-	-	25
Grand Total (INR)					2,829.20

Bill Drive Link: https://drive.google.com/drive/folders/1A4FhR7H-s3sZVWnyVxUpJtPzTDAFud6-?usp=sharing

Future scope

Advanced Vision & AI

Integration of YOLOv9 AI for improved obstacle classification.

Mapping & Navigation

SLAM-based path planning using LiDAR + Camera fusion for full-area coverage.
Room mapping with autonomous re-routing and real-time environment updates.

Sustainability & Adaptability

Use of eco-friendly materials and energy-efficient motors.
Modular design for easy sensor/motor upgrades and scalability.